

Table 3: Learning objectives (Theory) of Course

Paper 1 Ayurvediya Aushadhi Nirmana Vigyana									
A3 Course outcome	B3 Learning Objective (At the end of the session, the students should be able to)	C3 Domain/sub	D3 Must to know / desirable to know / Nice to know	E3 Level Does/ Shows how/ Knows how/ Know	F3 T-L method	G3 Assessment (Refer abbreviations)	H3 Formative/ summative	I3 Term	J3 Integration
Topic 1 1.Chronological development of Ayurvediya Aushadhi Nirmana (Lecture :2 hours, Non lecture: 1 hours)									
CO1	Explain historical evolution of Ayurvediya aushadhi nirman and Rasashastra.	CK	MK	K	L&G D	TT-Theory	F&S	I	
CO1	Describe about contribution of Nagarjuna Acharya to Rasashastra	CK	MK	K	L_ VC	TT-Theory	F&S	I	
CO1	Enlist important classical texts of Rasashastra and describe their unique features in short.	CK	DK	K	L&PP T	TT-Theory	F&S	I	
CO1	Describe structure of Pharmacy and enlist formulations prepared in pharmacy, after visiting the unit of teaching pharmacy of own campus	CC	MK	KH	L&G D	CL-PR	F	I	
CO1	Describe Recent development in Ayurvedic Pharmaceuticals viz, new dosage forms, pharmaceutical modification techniques.	CK	DK	K	L&PP T	TT-Theory	F&S	I	
CO1	Define Rasa and Rasayana and describe difference between Rasa and Rasayana	CK	DK	K	L&PP T	T- MEQs	F	I	

CO1	Justify design of ancient Rasashala	AFT- VAL	NK	KH	SDL	PRN	F	I	
Topic 2 2.Paribhasha (Terminology) (Lecture :8 hours, Non lecture: 4 hours)									
CO1	Explain the term Paribhasha and its importance in Ayurvediya Aushadhi Nirmana.	CK	MK	K	L	T-CRQs	F	I	
CO1	Discribe the terms Aushadha, Bheshaja, Kalpana, Kashaya, Kashaya yoni, Samskara- based on their word derivations	CC	MK	K	L	T- EW	F&S	I	
CO1	Enlist sequentially - names of all drugs classified in the varga(group). Recite shlokas of Maharasa, Uprasa , Sadharana Rasa Varga from Rasaratnasamuchchaya.	CC	MK	K	EDU, SDL, GBL, REC	P-REC,P-ID,PUZ,O-QZ	F&S	I	
CO1	Enlist and discuss Sandigdha, Krutrima, Pratinidhi and Anupalabdha dravya.	CK	DK	K	L&PP T,SD L,GB L	P-ID,CL-PR	F	I	
CO1,CO2	Discribe the definition of the term Shodhana. Explain with examples different techniques used for the procedures of Shodhana.	CAP	MK	KH	L&G D,L_ VC,P T	T- MEQs,P-E XAM,O-QZ,O-GAME	F&S	I	
CO1	Explain the term Marana and describe its types with examples	CK	MK	K	L&PP T,LS	T-OBT	F&S	I	
CO1	Enlist all relevent prakriya paribhasha of Amrutikarana, Lohitikarana,Sattvapatana, druti and discuss with examples	CK	DK	K	DIS,L S	PRN	F	I	
CO1,CO2	Recite sequentially names of Parada Ashta samskara. Explain the terms Jarana Murchchhana and cite types with examples. Compare Jarana and Murchchhana.	CC	MK	KH	L_ VC ,PrBL	T-EMI,T- ME Qs,PRN,M-CHT	F&S	I	

CO1	Identify names of Parada Bandhas	CK	NK	K	LS	T-OBT	F	I	
CO1,CO2,CO5	Describe Grahyaagrahya parameters used for selection of Rasadravyas.	CK	MK	SH	L&PP T,PrB L	T- EW	F&S	I	
CO1,CO2,CO5,CO6	Illustrate all Bhasma pariksha as per classical description.	CC	MK	KH	L_VC ,IBL, DA,D	T- MEQs,P- EXAM,CHK	S	III	
CO1,CO2,CO5	Define the word Siddhilakshani.Recite examples of classical siddhilakshani. Interpret its rationality	CE	MK	KH	L&PP T,SD L,RE C,D_ L	T- MEQs,P- EXAM	F&S	I	
CO1	Recognise and discuss Dhanvantari Bhaga and Rudra bhaga	CC	NK	K	RLE	C-INT	F	I	
CO1	Explain importance of Mana-paribhasha and classical types of Mana. Recall Charts of Mana .	CK	DK	KH	L&PP T,PS M	T-CRQs,P- SUR	F	I	
CO1,CO2	Categorize parameters of Drug Standardization and develop a checklist for assessment of quality of rasadravyas	CE	DK	KH	BS,IB L,TP W,SD L	CL- PR,WP,CHK	F&S	I	
CO2	Explain Value of selection of genuine raw material	AFT- VAL	MK	KH	DIS	DEB	F	II	
CO2,CO3	Explain importance of ethical practices for drug processing(Shodhan, Marana)	AFT- VAL	MK	K	PrBL	P-POS	F	II	
CO2,CO3	Discuss and justify importance of Bhasma Pariksha	AFT- RES	MK	KH	BS	T- EW	F&S	III	

CO2,CO3	Explain nalve of keen and accurate application of weights and measures in Ayurvediya Aushadhi nirmana	AFT-VAL	DK	K	L&G D	PRN	F	II	
Topic 3 3.Adharbhuta Siddhanta (Application of fundamental principles) (Lecture :4 hours, Non lecture: 2 hours)									
CO1,CO2	Elaborate fundamental principles of Ayurvediya Aushadhi Nirmana alongwith their classical references and discuss their application with classical examples of various kalpas.	CC	MK	KH	L&PP T,BS	T- EW,M-POS	F&S	I	
CO1,CO2	Explain Dravya Samgraha vidhi. Explain types of Bhumi desha and types of drugs to be collected from paricular place & places from where Dravya should not be collected.	CC	MK	KH	L&PP T,DIS ,BS	T- EW	F&S	I	H-DG
CO1,CO2	Describe time of Dravya collection and explain rationality behind it.	CC	MK	KH	L&PP T,BS	T- EW	F&S	I	H-DG
CO1,CO2	Enlist parts of Sthavara & Jangama Dravyas used for Ayurvediya Aushadhi nirmana.	CC	MK	KH	L&G D,BS	T- EW	F&S	I	H-DG
CO1,CO2	Define Samskara. Discuss role of Samskara in Ayurvediya Aushadhi Nirmana alongwith various examples.	CC	MK	KH	L&PP T,BS	T- EW	F&S	I	H-Sa mhita
CO1,CO2	Discribe Ardra - Shushka Dravya & Anukta – Visheshokta Dravya collection principles.	CC	MK	KH	L&PP T,DIS	T- EW	F&S	I	H-DG
CO1,CO2	Describe importance of kalpa sevan matra. Explain factors considered for deciding dosage of any drug (Ayurvedic as well as modern medicine principles). Describe Posology	CC	MK	KH	L&PP T,BS	T- EW	F&S	I	
CO1,CO2	Explain Saveeryata Avadhi(Shelf life) of Ayurvedic dosage forms.	CK	DK	K	L&PP T,DIS	T- MEQs,P-VIVA	F&S	I	
CO1,CO2	Discuss Yogika Dravya Sidhdhanta(Drug combination)	CC	MK	KH	L&PP T,BS	T- EW	F&S	I	
CO1	Explain importance of Kala (Time) Samskara in Ayurvediya	CK	MK	K	L&PP	TT-Theory	F&S	I	

	Aushadhi Nirmana. Elaborate Aushadhi sevana kala mentioned in Sharangdhara samhita. Discribe chrono- Therapeutics.				T				
CO1	Justify Aushadhi kalpa namakarana siddhanta with examples	CC	DK	KH	IBL	CL-PR	F	I	
Topic 4 4.Yantropakaranani- I (Equipments and machineries) (Lecture :6 hours, Non lecture: 4 hours)									
CO1,CO5	Choose and record contemporary machines used in Ayurvedic drug preparation.	CC	DK	K	L&PP T,DIS	T- EW	F&S	I	
CO1,CO5	Discuss the pharmaceutical use of Distillation apparatus, Ball Mill, Pulveriser, End Runner, Edge Runner, Tablet compression machine, Capsule filling machine, Pouch filling machine, Liquid filling machine in Ayurvediya Aushadhi Nirmana	CC	DK	K	L&PP T,DIS	T- EW	F	I	
CO1,CO5	Describe the principles behind construction and working of the classical Yantras used for Ayurvediya Aushadhi nirmana.	CAP	MK	KH	L&PP T,DIS	T- MEQs,P- VIVA	F&S	I	
CO1,CO5	Enlist categorical information about the following Yantras in the charts- Ulukhal Yantra, Patan Yantra, Jarana Yantra, Patala Yantra and Swedana Yantra.	CK	DK	K	L&PP T,DIS	T- EW	F	I	

CO1,CO5	Interpret the mechanism and effect of Yantras / machines on the Physical and Chemical properties of the drug material.	CC	DK	KH	DIS,I BL	M-POS	F	I	
CO1	Enlist Ayurvediya aushadhi kalpana and equipments/yantras/ machines used for preparation of each kalpana.	CK	DK	K	L&PP T	T- EW	F	I	
Topic 5 5.Yantropakaranani -II (Equipments, fuel and Heating Devices) (Lecture :5 hours, Non lecture: 4 hours)									
CO1,CO5	Describe the term Yantra and enlist yantras described in classics useful for aushadhi nirmana. -	CK	DK	K	L&PP T	TT-Theory	F	I	
CO1,CO5	Define the term Puta and recite its classical explanation.	CK	MK	K	L_VC ,RLE	T- EW,M- POS	F&S	I	
CO1,CO5	Produce categorical information(Size of Pit, Number of cowdunkakes, use etc.) about following Putas viz. Mahaputa, Gajaputa, Varahputa, Kukkutaputa, Kapotputa, Lavakputa, Kumbhaputa, Bhandaputa, Valukaputa and Bhudharputa.	CK	MK	K	L&PP T,DIS ,TPW	T- EW,M- POS,COM	F&S	I	
CO1,CO5	Recognise the principles behind construction and working of the classical Putas.	CC	MK	KH	L&PP T,DIS ,IBL	T- EW,M- POS	F&S	I	
CO1,CO5	Identify and enlist contemporary devices used in the preparation	CC	DK	K	L&PP	T- MEQs,QZ	F&S	I	

	of Bhasma.					T,DIS ,IBL	,M-POS				
CO1,CO5	Review the temperature pattern of various Putas by referring research articles.	CC	DK	K		TPW, SDL	QZ ,M-POS	F&S	I		
CO1,CO5	Compile names and significance of temperature measuring devices with reference to Puta.	CK	DK	K		DIS, ML	CL-PR,M- POS	F	I		
CO1,CO5	Enlist the specific Puta used for a particular Bhasma Nirmana.	CC	DK	KH		L&PP T	CL-PR,M- POS	F	I		
CO1,CO5	Enlist various devices used for heating during Ayurvediya Aushadhi nirmana.	CK	DK	K		L,DIS ,RLE, FV	T- EW,P-SUR	F&S	I		
CO1,CO5	Enlist constituents needed for Samanya Musha Nirmana.	CK	DK	K		L&PP T,IBL	T- MEQs	F&S	I		
CO1,CO5	Define the term Musha and describe uses of various types of Musha.	CK	DK	D		L_ VC ,D-M	T-OBT	F&S	I		
CO1,CO5	Enlist and discuss the material used for Sandhi Bandhan (while using Sharava and Kachakupi)	CK	NK	K		DIS,R LE	T-OBT	F	I		
CO1,CO5	Identify and record the types of Crucibles .	CK	NK	K		L&PP T	M-POS	F	I		

CO1,CO5	Interpret the effect of heat transformation in the material subjected to heating through Musha.	CAP	DK	KH	L&PP T,DIS ,D-M	PRN,CL-PR	F	I	
CO1,CO5	Recall Pakaj Utpatti Siddhant and interpret it for Agni Sannikarsha Sanskara.	CC	DK	KH	L&PP T,DIS	T- EW,CR- RED	F&S	I	
CO1,CO5	Explain the Pharmaceutical use of the Hot plate, heating mantle, induction stove, hot air oven, muffle furnace (horizontal and vertical type)	CC	NK	KH	DIS,S DL	P-SUR,M- POS	F	I	
CO1	Describe types of Koshthi and their uses	CK	DK	K	L	COM	F	I	
CO1,CO5	Describe the Current trends in heat transfer device e.g. steam jacketed heating device, programmed muffle furnace, programmed furnace for Parpati / Pottali preparation.	CC	DK	K	L&PP T,IBL ,SDL	M-POS,COM	F	I	
CO1,CO5	Assess and interpret the effect of fuel in quantum of heat given (time and temperature)	CC	DK	KH	DIS,P rBL	M-POS,CR- RED	F	I	
CO1,CO5	Enlist various fuels used for heating for estimation of their heat value.	CK	NK	K	IBL,S DL	M-POS	F	I	
CO2,CO3	Analyse and appraise use of proper heating device and fuels used for Ayurvediya Aushadhi Nirmana.	AFT- SET	DK	KH	BS	DEB	F	II	
Topic 6 6.Kalpana Nirmana I (Primary & Secondary dosage forms) (Lecture :6 hours, Non lecture: 4 hours)									
CO1,CO2	1. Explain Kashaya kalpana 2. Enlist the dosage form come under Panchavidha Kashaya	CC	MK	KH	L	T- EW,P- VIVA	F&S	II	

	-Ingredients, proportion, matra, sevan vidhi if any, therapeutic uses												
CO2	Critically evaluate rationality behind different proportion of water used in various formulations	AFT-VAL	DK	KH	BS	P-VIVA	F&S	III					
Topic 7 7.Kalpana Nirmana-II (Method of Preparation of different dosage forms& Dietary Supplements)) (Lecture :5 hours, Non lecture: 4 hours)													
CO1,CO5	Explain the basic principles involved , processing techniques,quality control parameters used/ involved in the preparation of Avaleha, Kalpana with examples ,instruments used in small and Large scale production,Research updates	CAP	MK	K	L&PP T,TP W,BL	T- EW,T-OBT	F&S	II					
CO1,CO5	Explain the basic principles involved , processing techniques,quality control parameters used/ involved in the preparation of Sneha Kalpana, Kalpana with examples ,instruments used in small and Large scale production,Concept of Avartana, Research updates on Snehakalpana, Market survey	CK	MK	K	L_ VC ,TUT	T- EW,P- VIVA	F&S	II					
CO1,CO4	Describe the Significance of sandhana kalpana, classification, Difference between Madya and Shukta Kalpana, general method of preparation, essential ingredients, Anukta Mana, Sandhana Vidhi, observations, Burnig candle test, Lime water Test, difference between Asava and Arishta, essential knowledge of sale and clinical practice , Research updates	CAP	MK	K	L_ VC ,PrBL	T- EW,T-OBT	F&S	II					
CO1,CO3,CO5	Explain Definition , significance of Pathya, types, general, method of preparation Manda, Peya, Yavagu, Vilepi, Anna or Odana Kalpana, Krushara, Yusha, Takra, Khada, Kambalika, Raga, Shadava, Related Research updates, Market survey of Dietary Supplements and Nutraceuticals	CAP	MK	KH	L_ VC ,RLE	T- EW,P-VIV A,P-EXAM	F&S	II					
Topic 8 8.Rasa Dravya Parichaya- I (Lecture :12 hours, Non lecture: 4 hours)													
CO1		CC	MK	K	L&PP	T- MEQs,P-	S	I					

	Enlist synonyms of Rasadravya and explain significance of that					T,D_ L	VIVA				
CO1,CO2,CO 6	Explain classical & Mineralogical identification and Sources of Rasadravyas	CC	MK	SH		L&PP T,PT	T- EW,P- EXAM	F&S	I		
CO1,CO2	Discriminate types, Grahya-Agrahyata and Dosha of Rasadravya	CC	MK	KH		L&PP T,PT	P-VIVA,TT- Theory	F&S	I		
CO1,CO2	Explain Shodhan and Marana and other processing techniques of the Rasa- Dravya.	CC	MK	KH		L_V C ,IBL, D	T- EW,P- VIVA,INT	F&S	I		
CO1,CO2,CO 6	Discuss probable physical, chemical changes occurring during process & after Shodhana and Marana of Parada, Haratala, Tamra and Shankha.	CC	MK	KH		L&PP T,BS, SDL	T- EW,P-VIV A,P-EXAM	F&S	I		
CO6		CC	DK	KH		DIS,B	P-VIVA	F&S	I		

	Explain research updates about Shilajatu,Naga,Mukta and Kapardika.?					S					
CO1,CO2,CO6	Discuss probable physical, chemical changes occurring after Bhavana, Puta & whole process of Bhasma nirmana from Rasadravyas	CC	MK	KH	DIS,B S	T-OBT	F&S	I			
CO1,CO4	Enlist Kalpas prepared from these Rasdravyas and therapeutic importance of that Rasadravya	CK	DK	K	IBL,S DL	PRN	F	II			
CO2,CO3	Develop checklist for identification of genuine rasadravyas	AFT- CHR	MK	K	IBL	P-PS	F&S	III			
Topic 9 9.Rasa Dravya Parichaya II (Lecture :7 hours, Non lecture: 6 hours)											
CO1	Enlist the important synonyms of Rasadravyas	CC	DK	KH	L&PP T,DIS	P-VIVA	F&S	II			
CO1,CO6	Explain classical & Mineralogical identification and their Sources.	CK	DK	K	L&PP T,D_ L	P-VIVA	F&S	II			

CO1,CO6	Determine types, Grahya-Agrahya, properties & classical as well as Mineralogical Identification of Rasadravyas	CC	NK	K	L&PP T,DIS ,SDL	P-VIVA,CHK	F&S	II		
CO1,CO2	Explain Shodhana, Marana and Probable Chemical Changes.	CC	DK	K	L&PP T,IBL	PRN	F&S	II		
CO6	Explain research update of Badarashma.	CK	NK	K	DIS,L S	P-VIVA	F	II		
CO1,CO5	Enlist Rasadravyas from this group and describe important kalpas with their therapeutic utility	CK	DK	K	TPW	PRN	F	II		
Topic 11 11.Kalpana Nirman -III (Method of Preparation of different dosage forms) (Lecture :8 hours, Non lecture: 6 hours)										
CO1,CO2,CO 3	Describe sharkara kalpana along with preparation method of Syrup.Explain therapeutic importance of prepared sharkara kalpana along with its shelf life.	CC	NK	K	L&PP T,SD L	P-VIVA	F&S	II		
CO1,CO2,CO 3	Describe Gudapaka preparation method with its confirmatory tests and precautionsExplain therapeutic importance of prepared Gudapaka along with its shelf life.	CC	NK	K	L&PP T	P-VIVA	F&S	II		
CO1,CO2,CO 3	Describe Lavana kalpana preparation method with its confirmatory tests and precautionsExplain therapeutic	CC	DK	K	L&PP T	T-CRQs,P- VIVA	F&S	II		

	importance of prepared Lavana kalpana along with its shelf life and packaging techniques.												
CO1,CO2,CO3	Describe Kshara kalpana preparation method with its confirmatory tests and precautionsExplain therapeutic importance of prepared Kshara kalpana and Ksharasutra integration with Shalya tantra department along with its shelf life and packaging techniques.	CAP	MK	KH	L&PP T	P-VIVA,TT-Theory	F&S	II		V-SHL			
CO1,CO2,CO3	Describe Ayaskriti kalpana preparation method with its confirmatory tests and precautionsExplain therapeutic importance of prepared Ayaskriti kalpana along with its shelf life and packaging techniques.	CC	NK	KH	L&PP T	P-VIVA	S	II					
CO1,CO2,CO3	Describe Lepa kalpana preparation method with its confirmatory tests and precautionsExplain importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	CC	MK	K	L&PP T	T-EW,P-VIVA	F&S	II		V-KC,V-PC			
Topic 12 12.Chaturvidha Rasayana (Lecture :4 hours, Non lecture: 4 hours)													
CO1,CO2	Describe importance and unouqe features of Chaturvidha Rasayana - Kharaliya Rasayana, Parpati Rasayana, Kupipakwa Rasayana & Pottali Rasayana.	CC	MK	KH	L&PP T,IBL	T-EW,P-VIVA	F&S	II					
CO1,CO2	Explain definition, types and method of preparation of Chaturvidha Rasayana.	CAP	MK	D	L&PP T,PT	T-EW	F&S	II					

CO1,CO2	Determine role of Yantra, Agni & Kala for pharmaceutical process of chaturvidha rasayana	CAN	DK	KH	PT	T- EW,P- VIVA	F&S	II	
CO6	Explain shelf-life of Chaturvidha Rasayana.	CAN	DK	SH	L,DIS	P-VIVA	F&S	II	
CO5	Describe chemical changes occurring during the preparation of chaturvidha rasayana and its impact on Pharmacological action of Chaturvidha Rasayana	CAN	DK	KH	L&G D,BS, SDL	CR-RED,CR- W	F	II	
CO5	Explain law of Definite proportion and proportion of Mercury and sulphur needed for preparation of sulphide of Mercury. in Chaturvidha rasayana	CAP	DK	KH	BS,IB L	CL-PR	F	II	
Topic 13 13.Current and emerging trend in Ayurvedic pharmaceuticals (Lecture :3 hours, Non lecture: 4 hours)									
CO1,CO5,CO 6	Discuss the classification of different dosage forms.	CK	DK	K	L&PP T,DIS	P-VIVA,M- POS	F&S	III	
CO1,CO5,CO 6	Explain the need of different dosage forms.	CK	DK	K	L&PP T,DIS ,SDL	P-VIVA,M- POS	F&S	III	
CO1,CO5,CO	Enlist categorical information about the cosmetics used in daily	CK	DK	K	L VC	PRN,M-POS	F	III	

6	routine. Appreciate Ayurvedic cosmetic products.												
CO1,CO5	Appreciate Ayurvedic cosmetic products.	AFT-VAL	DK	KH	BS,Pr BL	PRN,P-SUR	F	III					
CO1,CO5,CO6	Choose and record the contemporary machineries used in the manufacture of cosmetics.	CK	NK	K	L_VC ,DIS, SDL	M-POS	F	III					
CO1,CO5,CO6	Enlist Quality Control parameters of cosmetics preparations.	CK	DK	K	L&PP T,DIS	P-VIVA,M-POS	F&S	III					
Topic 14 14.GMP(Schedule T) & Regulatory aspects of Ayurvedic drugs (Lecture :2 hours, Non lecture: 4 hours)													
CO5	Explain the legal and regulatory aspects of manufacturing, and sale of Ayurvedic drugs.	CC	MK	K	L&PP T,IBL	CL-PR	F&S	III					
CO5	Describe acts and rules mentioned in Drug & Cosmetic Act 1940 & Rule 1945 and their relevance to Ayurvedic, Siddha, Unani (ASU) drugs.	CC	MK	K	L&G D,BS	QZ	F&S	III	H-DG				
CO5	Discuss the guidelines of Food Safety and Standards Authority of India (FSSAI) and FDA.	CK	DK	K	L&G D,SD L	CL-PR	F	III					
CO3,CO5	Determine the principles and practice of establishment of Ayurvedic pharmacy.	CK	DK	K	L&G D,TP W	P-EN,CL-PR	F	III					
CO5	Discuss the NABL guidelines for testing laboratory (Chemical).	CK	NK	K	L&PP T,D_	CL-PR,CHK	F&S	III					

CO1,CO2,CO4,CO5,CO6	Describe therapeutic efficacy of different formulations of Vishadravya (Bhallataka), with its toxic effects and remedy.	CK	NK	K	L&G D	P-VIVA	F	I	
CO1,CO5	Describe therapeutic efficacy of different formulations of Guduchi, Gairika, Gandhaka	CC	MK	K	L&G D,PE R	P-VIVA,TT- Theory	F&S	III	
CO3,CO4	Appraise multiple factors considered for preparation of various dosage forms from a single drug and creat list of more such examples.	AFT- CHR	DK	KH	BS,IB L	COM	F	III	
Topic 3 3.Single drug(Bhasma, Shuddha & Pishti) (Lecture :12 hours, Non lecture: 6 hours)									
CO2,CO6	Explain chemical form/composition, Pharmacodynamics and pharmacokinetics of Abhraka, Loha and Godanti Bhasma.	CC	DK	KH	L&PP T,DIS ,IBL	P-VIVA	S	III	
CO1,CO2	Explain therapeutic properties, dosage, Anupana, Pathyapathya, duration of treatment, Sevana Kala, shelf life, important Yogas of each Bhasma.	CC	MK	KH	L&PP T,TP W	T- EW,P- VIVA	F&S	III	
CO1,CO2	Describe Apakwa Ashuddha Avidhi Bhasma Sevanajanya Vyadhi and their Shanti-upaya.	CK	DK	KH	DIS,B S	P-VIVA,TT- Theory	F	III	
CO1,CO2	Describe in detail Amayika Prayoga (Therapeutic uses) of each Bhasma/Pishti with given references.	CAP	MK	SH	L&G D,W	T- EW,P- VIVA	F&S	III	

CO3,CO4	Explain ethical aspect of administration of compound drugs mainly regarding duration of the treatment given.	AFT-VAL	DK	K	PBL	CL-PR	F	III	
CO1,CO2,CO4,CO6	Describe ingredients, with its dose, therapeutic importance, anupana and pathya-apathya along with its clinical evidences and Research upadates	CK	MK	KH	L&PP T	T- EW,P- VIVA	F&S	III	
CO1,CO2,CO4,CO6	Enlist the formulations with its dose, anupana and indications	CK	NK	K	L	T-EMI	S	III	
Topic 6 6.Dosage Forms & Cosmetic Products (Lecture :5 hours, Non lecture: 2 hours)									
CO2,CO3,CO7	Define the term- Dosage forms.	CK	MK	K	L&PP T	TT-Theory	F&S	III	
CO2,CO3,CO7	Discuss the need of different dosage forms.	CK	MK	K	L&PP T	TT-Theory	F&S	III	
CO2,CO3,CO7	Explain the classification of different dosage forms.	CK	DK	K	L&PP T	TT-Theory	F&S	III	
CO2,CO3,CO7	Enlist the solid / liquid / semisolid dosage forms.	CK	MK	K	L&PP T	T-EMI	F&S	III	
CO2,CO3,CO7	Discuss the routes of administration of different dosage forms.	CK	MK	K	L&PP T	T- EW	F&S	III	
CO2,CO3,CO7	Describe the advantages and disadvantages of currently available dosage forms.	CK	MK	K	L&PP T	TT-Theory	F&S	III	
CO2,CO3,CO7	Discuss the research updates about modification of classical Ayurvedic dosage forms.	CK	DK	K	L&PP T	CL-PR	F	III	
CO2,CO3,CO7	Discuss the relevant case studies of different dosage forms.	CK	NK	K	L&PP T	P-SUR	F	III	

CO2,CO3,CO7	Define the term Cosmetics.	CK	DK	K	L	TT-Theory	F&S	III	
CO2,CO3,CO7	Explain the classification of cosmetics based on their application on the specific body parts.	CK	DK	K	L&PP T	CL-PR	F&S	III	
CO2,CO3,CO7	Differentiate between the ancient cosmetics and modern day cosmetics.	CK	DK	K	L&PP T	TT-Theory	F&S	II	
CO5	compare and appraise herbal cosmetics vs synthetic products	AFT- VAL	DK	KH	PBL	PRN	F	III	
Topic 7 7.Nutraceuticals (Lecture :6 hours, Non lecture: 1 hours)									
CO1	Describe importance of combination of aahara and aushadha	CK	MK	K	L&PP T	T- EW	F&S	III	H-SW
CO2,CO6	Explain Nutraceuticals with its types	CK	MK	K	L&PP T	TT-Theory	F&S	III	H-SW
CO2,CO6	Corelate rasayana and nutraceutical with examples	CAN	DK	KH	DIS,P rBL	T-OBT	F	III	H-SW
CO2	Describe with examples mode of action of Ayurvedic nutraceuticals	CAP	MK	KH	SDL, LS	PRN	F	III	H-Sa mhita, H-SW
CO2,CO6,CO7	Enlist at least two dietary preparations from Ayurvedic classics which can serve as rasayana/ nutraceutical in-1) General Health2) Pediatric Health3) Geriatric Health4) Women(Garbhini/ Sutika) Health5) Cardio-protection6) Chronic illness- recovery stage	CAP	MK	K	DIS,B S,Sy	CL-PR	F	III	H-SW
CO1,CO2,CO6,CO7	Describe potential of Ayurvedic diet and Rasayana in sports endeavor	CAP	DK	KH	BS,Pr BL,PS	DEB	F	III	H-SW

CO2,CO4	Enlist the dosage forms used in the specific Aushadhi Prayog Marga.	CK	DK	K	L&PP T	T-EMI	F&S	II	
CO2	Discuss the nature of drug (s) administered in various routes of drug administration.	CAP	NK	KH	L&G D	CL-PR	F	II	
CO4,CO5	Appraise administration of drugs through various routes mentioned in Ayurvedic treatment	AFT- VAL	MK	K	L&G D	T- EW	F&S	III	
Topic 10 10.Rational prescription along with safe dispensing of Ayurvedic formulations. (Lecture :1 hours, Non lecture: 4 hours)									
CO2,CO4	Describe and write demo ideal prescription.	CAP	MK	SH	L&G D,CB L	CR-W,CHK	F&S	III	V-KC V-SP V-BL
CO1,CO3,CO 4	Explain the safe dispensing and efficacious use of Ayurvedic drugs.	CAP	MK	KH	L&G D,BS, TPW	T-CS,PM	F&S	III	V-KC V-BL
CO2,CO4	Explain the importance of rational prescribing of drugs and the concept of essential drugs.	CK	DK	K	L&G D,IBL	QZ ,CHK	F	III	V-KC
CO2,CO4,CO 5	Describe the standard protocol for safe dispensing of Ayurvedic drugs.	CK	DK	KH	L&G D	T-OBT	F	III	
CO2,CO4	Demonstrate and educate home remedies to small group of population.	CAP	DK	SH	PrBL, TPW	TR	F	III	H-SW
CO5	explain ethical aspects related to prescription writing	AFT- RES	MK	KH	TPW	INT	F	III	
Topic 11 11.Traditional & Local health Practices (Lecture :2 hours, Non lecture: 4 hours)									
CO1	Identify Local Health Traditions and Healing Knowledge,	CC	DK	K	DIS,P rBL	P-SUR,INT	F	III	

CO1,CO5	Recognise the possible potential of product development and research based on Traditional knowledge	CK	DK	KH	IBL	INT	F	III	
CO1,CO4	Identify the factors responsible for grant of patent and erroneous grant of patent on indian traditional knowledge.?	CK	DK	K	BS	INT	F&S	III	
CO5	Appraise traditional knowlege of Ayurvedic medicines	AFT-VAL	DK	K	TPW	CR-W	F	III	
Topic 12 12.Pharmacovigilance for Ayurveda drugs (Lecture :1 hours, Non lecture: 4 hours)									
CO2,CO4,CO5	Describe the term Pharmacovigilance and explain importance of Pharmacovigilance for Ayurvedic drugs.	CK	DK	K	L&PP T,IBL	M-POS,C- INT,RK	F	III	V-KC ,V-BL ,H-D ,G,H- AT
CO4,CO5	Explain the status and central sector scheme of Pharmacovigilance for Ayurveda, Siddha, Unani, and Homeopathy (ASU & H) drugs.	CK	MK	K	L&PP T,DIS	INT,TT- Theory	F	III	V-KC ,H-D ,G,H- AT
CO4,CO5	Define Adverse Drug Reactions (ADR) and its types.	CE	DK	K	L&G D,IBL	QZ,TT-Theor y,VV-Viva	S	III	V-KC ,V-BL ,H-D ,G,H- AT
CO2,CO4,CO5	Identify and monitor ADRs.	AFT-VAL	DK	KH	L&G D,BS, IBL	PRN,RK	F	III	V-KC ,V-BL ,H-D ,G,H- AT

CO2,CO4	Discuss and make critical comments on the safe and efficacious use of Ayurvedic drugs.	CAP	DK	KH	L&G D,PrB L,TU T	QZ ,CL- PR,INT	F	III	V-KC ,H-D G,H- AT
CO4,CO5	Debate on ADR of ASU drugs	AFT- RES	DK	K	DIS	DEB	F	III	